

Woxsen University

School of Technology



PROJECT REPORT

on

AI – POWERED AVIATION ANALYTICS

Aircraft Engine Predictive Maintenance using Machine Learning

Submitted in partial fulfilment of the requirements for the degree of

B. Tech. in Artificial Intelligence and Machine Learning

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DECLARATION OF THE CANDIDATE

We hereby declare that the project work entitled "AI-Powered Aviation Analytics —Aircraft Engine Predictive Maintenance using Machine Learning" submitted to the School of Technology, Woxsen University, in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning is our original work and has been carried out under the guidance of Geeta Tripathi.

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma at this or any other institute or university

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ACKNOWLEDGMENT

We would like to express our sincere gratitude to all those who contributed to the successful completion of this project on AI-Powered Aviation Analytics and aircraft engine predictive maintenance.

First and foremost, we thank our project mentor Geeta Tripathi, professor of learning and practice, for providing guidance, feedback, and academic direction throughout the development of the project

We are grateful to Woxsen University and the School of Technology for providing the academic environment and resources required to carry out this work.

We also acknowledge the availability of the NASA C-MAPSS FD001 benchmark dataset, which provided the sensor-based run-to-failure data used for the predictive-maintenance study.

We thank our peers and project teammates for their discussions, testing, and feedback during the implementation and evaluation stages.

Finally, we express our gratitude to our families for their continued support and encouragement throughout the project.

ABSTRACT

This project presents an academic predictive-maintenance system for estimating the Remaining Useful Life (RUL) of aircraft engines from multivariate sensor measurements. The study uses the NASA C-MAPSS FD001 benchmark dataset and develops a machine learning workflow that converts sequential engine operating data into RUL predictions.

The implemented pipeline includes sensor inspection and removal of constant sensors, feature normalization using Standard Scaler, generation of RUL targets from engine failure cycles with an upper cap of 125 cycles, five-cycle sensor smoothing, and ten-cycle rolling statistical and slope features for the tabular Random Forest model. For sequence modelling, the project constructs 30-cycle LSTM input windows.

Two regression approaches are evaluated: Random Forest as a tabular baseline and a Long Short-Term Memory (LSTM) neural network as the primary sequence-based model. The trained models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the PHM08 scoring function. On the documented evaluation of 100 unseen test engines, Random Forest obtained an MAE of 13.63, RMSE of 18.82, and PHM08 score of 662.92, while the LSTM obtained an MAE of 10.63, RMSE of 15.24, and PHM08 score of 355.81.

The trained models are integrated into a Streamlit dashboard that provides fleet-level and engine-level views, predicted RUL, project-defined maintenance status categories, sensor trends, benchmark actual RUL, prediction error, and model comparison. The system is explicitly an academic/conceptual demonstration using the NASA benchmark dataset; it does not use live aircraft data and is not a certified aviation maintenance system.

Keywords: Predictive Maintenance, Remaining Useful Life, Aircraft Engine, NASA CMAPSS FD001, Random Forest, LSTM, Machine Learning, Time-Series Sensor Data, Streamlit.

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1.INTRODUCTION

Aircraft engines operate under changing loads and environmental conditions, and their sensor measurements can provide information about the progression of degradation. Predictive maintenance attempts to use such measurements to estimate the remaining useful life of equipment before a failure endpoint. In this project, the task is formulated as a supervised regression problem in which historical engine sensor measurements are used to estimate the number of operating cycles remaining.

1.1 Background

The project focuses on Remaining Useful Life (RUL) prediction using the NASA C-MAPSS FD001 benchmark dataset. Each engine is represented by a sequence of operating cycles with multiple sensor measurements and operating settings. The training data contain complete run to-failure histories, allowing an RUL target to be constructed from the final observed cycle of each engine. The project combines a conventional tabular regression model and a sequence-based neural network. Random Forest is used as a baseline because it can operate on engineered tabular features. LSTM is used as the primary model because it is designed to learn patterns across ordered sequences of observations.

1.2 Motivation

- Estimate Remaining Useful Life from multivariate engine sensor measurements.
- Compare a conventional machine-learning baseline with a sequence-based neural model.
- Demonstrate a complete workflow from preprocessing and feature engineering to model evaluation and dashboard inference.
- Provide interpretable project-level maintenance status categories based on predicted RUL.
- Create a reproducible academic implementation using a public benchmark rather than claiming access to live aircraft data.

1.3 Problem Statement

The primary problem addressed by this project is the estimation of the Remaining Useful Life of aircraft engines from time-series sensor data. The challenge is to convert noisy, multivariate and temporally ordered measurements into a numerical estimate of remaining operating cycles. The system must also avoid misleading claims about real-world aviation maintenance because the evaluation is performed on a benchmark dataset rather than operational aircraft data.

1.4 Objectives

1. Load and process the NASA C-MAPSS FD001 training, test, and RUL files.
2. Identify and remove constant sensors and normalize the remaining input features
3. Generate RUL targets from engine failure cycles and cap the training target at 125 cycles.
4. Create smoothed and rolling statistical features for Random Forest regression.
5. Create 30-cycle sequences for LSTM-based RUL prediction.
6. Train and evaluate Random Forest and LSTM approaches.
7. Compare the models using MAE, RMSE, and PHM08 scoring.
8. Integrate the saved models into a Streamlit dashboard for engine-level analysis.

1.5 Dataset Overview

The repository contains the NASA C-MAPSS FD001 files `train_FD001.txt`, `test_FD001.txt`, and `RUL_FD001.txt`. The implementation names each record by engine unit and operating cycle, followed by three operating settings and twenty-one sensor columns in the raw schema. Constant sensors are subsequently identified from the training data and removed before normalization.

Attribute	Project Implementation
Dataset	NASA C-MAPSS FD001
Training File	<code>data/raw/train_FD001.txt</code>
Test File	<code>data/raw/test_FD001.txt</code>
RUL File	<code>data/raw/RUL_FD001.txt</code>

Table 1.1: NASA C-MAPSS FD001 Project Dataset Characteristics

1.6 Scope and Limitations

- The project is limited to the NASA C-MAPSS FD001 benchmark dataset and its sensor configuration.
- The dashboard uses saved trained models and does not retrain them at startup.
- The maintenance status thresholds are project-defined interpretation rules and are not official aviation maintenance limits.
- The project does not use live aircraft data and therefore cannot establish operational aviation performance.
- Generalization to another aircraft type, sensor configuration, or real-world fleet would require compatibility checks and potentially additional training or adaptation.

1.7 Organization of the Report

Chapter 2 reviews predictive maintenance, RUL estimation, regression models, sequence modelling, and evaluation metrics relevant to the project. Chapter 3 explains the implemented methodology. Chapter 4 describes the software implementation and dashboard. Chapter 5 presents the documented results and their interpretation. Chapter 6 describes testing and validation. Chapter 7 discusses contribution, limitations, and future scope. Chapter 8 concludes the report and summarizes the key learnings.

2. LITERATURE REVIEW

2.1 Predictive Maintenance and RUL Estimation

Predictive maintenance uses equipment condition data to plan maintenance. RUL estimation predicts the remaining operating cycles before failure. The C-MAPSS dataset is widely used for evaluating RUL prediction methods.

2.2 Aircraft Engine Health Monitoring

Aircraft engines generate sensor measurements over multiple operating cycles. The main challenge is to identify degradation progression from these sensor readings. C-MAPSS provides a controlled benchmark for testing such methods.

2.3 Machine Learning for RUL Prediction

In this project, rolling mean, standard deviation, and slope features are extracted from sensor data. These features are used with Random Forest regression to capture recent degradation behaviour.

2.4 Random Forest for Regression

Random Forest combines multiple decision trees for regression. It is used as the baseline model with 100 trees and a random state of 42. GridSearchCV is also used for hyperparameter tuning.

2.5 LSTM for Sequential Sensor Data

LSTM networks learn patterns from sequential sensor data. The project uses sequences of 30 operating cycles, an LSTM layer with 64 units, and a Dense output layer for RUL prediction.

2.6 Evaluation Metrics

The models are evaluated using **MAE, RMSE, and PHM08 score**. MAE measures average prediction error, RMSE gives more weight to large errors, and PHM08 provides an asymmetric prognostic score. **Lower values indicate better performance.**

Metric	Interpretation
MAE	Average absolute prediction error
RMSE	Penalizes larger prediction errors more

Metric	Interpretation
PHM08 Score	Asymmetric prognostic evaluation score

Table 2.1: Evaluation Metrics Used in the Project

2.7 Research/Project Gap

The project integrates **data processing, RUL estimation, Random Forest and LSTM modelling, performance evaluation, and dashboard-based prediction** into one workflow using the C-MAPSS dataset. It focuses on academic benchmarking rather than real-time aircraft deployment.

3.METHODOLOGY

3.1 System Overview

The proposed workflow follows: **NASA C-MAPSS FD001 → Data Loading → Preprocessing → RUL Generation → Feature Engineering → Random Forest & LSTM Modelling → Evaluation → Dashboard Inference.**

3.2 Dataset and Preprocessing

The FD001 training, test, and RUL files are loaded using Pandas. The data contains engine **unit IDs, operating cycles, 3 settings, and 21 sensor measurements**. Constant and low-variance sensors are removed to eliminate non-informative features. The remaining features are normalized using **StandardScaler**, fitted only on the training data.

3.3 RUL Generation and Feature Engineering

For each engine, the maximum recorded cycle is considered the failure point. RUL is calculated as:

$$\text{RUL} = \text{Failure Cycle} - \text{Current Cycle}$$

and capped at **125 cycles**. Sensor data is smoothed using a **5-cycle rolling mean**. For Random Forest, **10-cycle rolling mean, standard deviation, and slope** are generated to capture local degradation patterns.

3.4 LSTM Sequence Construction

The data is arranged chronologically for each engine. Sliding windows of **30 consecutive cycles** are created, with the RUL of the final cycle used as the target. This provides temporal information to the LSTM model.

3.5 Model Development

Random Forest: Uses 100 trees with `random_state=42` and parallel processing. GridSearchCV with three-fold cross-validation is also available for tuning.

LSTM: Uses **30 time steps × 18 features**, an LSTM layer with **64 units**, and a Dense(1) output for RUL prediction. Adam optimizer, batch size **64**, maximum **20 epochs**, and early stopping with patience **3** are used.

3.6 Training and Evaluation

Engine IDs are split into **80% training and 20% validation** using `random_state=42`, preventing data leakage between sequences of the same engine. Models are evaluated using **MAE, RMSE, and PHM08 score**, where lower values indicate better performance.

3.7 Dashboard Integration

The Streamlit dashboard loads the trained Random Forest and LSTM models without retraining. It uses the latest **30 cycles** for LSTM prediction and displays **predicted RUL, maintenance status, sensor trends, actual RUL, prediction error, and model performance**.

Component	Configuration
RUL Cap	125 cycles
Sensor Smoothing	5-cycle rolling mean
RF Features	10-cycle mean/std/slope
LSTM Sequence	30 cycles $\times$ 18 features
RF Trees	100
LSTM Units	64
Batch Size	64
Max Epochs	20
Validation Split	20% engines
Evaluation	MAE, RMSE, PHM08

Table 3.1: Key Configuration

4. IMPLEMENTATION

4.1 Project Structure

The project is organized into separate modules for data, preprocessing, modelling, evaluation, and dashboard deployment.

Path	Purpose
data/raw/	NASA C-MAPSS FD001 dataset
models/	Saved Random Forest and LSTM models
src/preprocessing.py	Data cleaning, normalization and feature engineering
src/models.py	Random Forest and LSTM model functions
src/evaluate.py	MAE, RMSE and PHM08 evaluation
src/losses.py	PHM08-inspired loss and scoring
dashboard/app.py	Streamlit dashboard and inference

Table 4.1: Project Structure and Purpose

4.2 Software and Technology Stack

The implementation uses Python as the primary language. Pandas and NumPy are used for data processing, while Scikit-learn provides preprocessing, Random Forest, data splitting, and hyperparameter tuning. TensorFlow/Keras is used for LSTM modelling. Streamlit and Plotly support the interactive dashboard, and Joblib is used for Random Forest model serialization.

Technology	Purpose
Python	Core implementation
Pandas, NumPy	Data processing
Scikit-learn	Preprocessing and Random Forest
TensorFlow/Keras	LSTM modelling
Plotly, Streamlit	Dashboard and visualization

Table 4.2: Software and Technology Stack

4.3 Model Files and Inference

The trained models are stored as `random_forest.pkl` and `lstm_final.keras`. The dashboard loads these models at runtime without retraining. LSTM is used as the primary prediction model, while Random Forest provides a baseline for comparison.

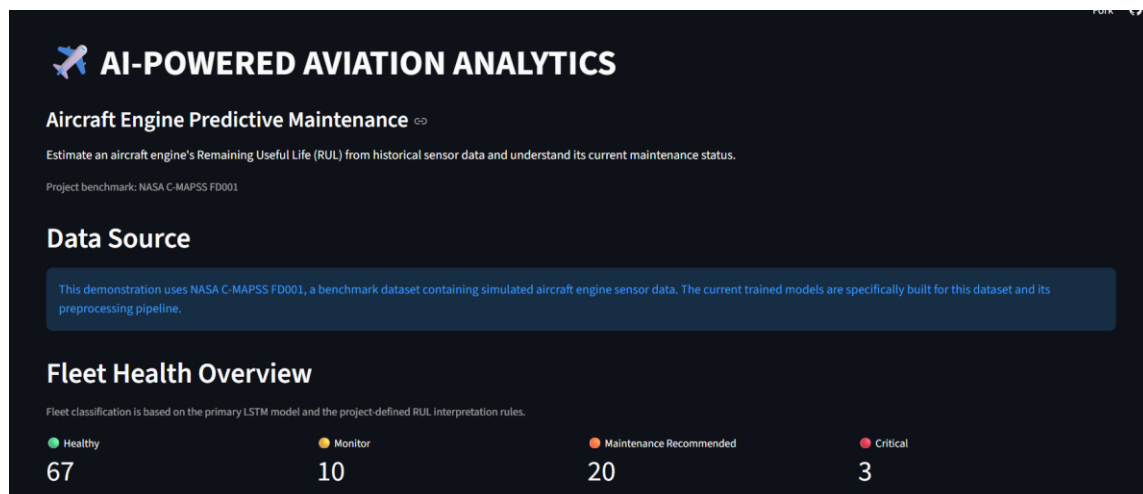
4.4 Streamlit Dashboard

The Streamlit dashboard provides both fleet-level and engine-level analysis. It displays the health status of 100 test engines, predicted RUL, maintenance status, sensor trends, actual RUL, prediction error, and Random Forest–LSTM comparison.

4.5 Maintenance Status Interpretation

Predicted RUL	Status
> 50 cycles	Healthy
30–50 cycles	Monitor
10–30 cycles	Maintenance Recommended
< 10 cycles	Critical

Table 4.3: Project-Defined Maintenance Status Interpretation



5. RESULTS AND DISCUSSION

5.1 Model Performance Comparison

Both models were evaluated on **100 unseen FD001 test engines**. The LSTM achieved better performance than the Random Forest model across all three-evaluation metrics.

Model	MAE	RMSE	PHM08
Random Forest	13.63	18.82	662.92
LSTM	10.63	15.24	355.81

Table 5.1: Model Performance Comparison

5.2 Results Interpretation

The LSTM achieved an **MAE of 10.63**, compared with 13.63 for Random Forest, indicating lower average prediction error. Its **RMSE of 15.24** was also lower than Random Forest's 18.82. The **PHM08 score of 355.81** was substantially lower than 662.92, where lower values indicate better performance. These results support using **LSTM as the primary dashboard model** for the FD001 dataset.

5.3 Engine-Level Prediction

For an individual engine, the dashboard uses its **latest 30 operating cycles** as input to the saved LSTM model. It displays the predicted RUL along with the benchmark actual RUL and prediction error.

5.4 Dashboard Outputs

The dashboard provides:

- Fleet health overview of **100 test engines**
- Engine-level predicted RUL and latest cycle
- Project-defined maintenance status
- Sensor trend visualization
- Actual RUL and prediction error
- Random Forest vs LSTM comparison
- Actual vs predicted RUL visualization

5.5 Discussion

The results indicate that **temporal sequence modelling using a 30-cycle LSTM** performs better than the Random Forest baseline for this FD001 evaluation. Random Forest remains useful as a simpler baseline, while the LSTM better captures sequential degradation patterns. These results are specific to the benchmark evaluation and do not guarantee performance in real-world aircraft operations.



6. TESTING AND VALIDATION

6.1 Pipeline Validation

The preprocessing pipeline includes validation checks for missing values, RUL range, train/test column consistency, engine-level splitting, and LSTM sequence dimensions.

Validation Check	Expected / Implemented Condition
Missing Values	Train and test missing-value counts checked
RUL Range	RUL remains between 0 and 125 after clipping
Train/Test Columns	Input feature columns are checked for consistency
LSTM Sequence	Sequence contains 30-time steps
Engine Split	Training and validation engines are separated

Table 6.1: Pipeline Validation Checks

6.2 Data Validation

The preprocessing stage checks sensor variance, identifies constant sensors, verifies train/test feature consistency, and reports data dimensions. These checks help identify data-related issues before model training.

6.3 Model Validation

Unique engine IDs are divided into **80% training and 20% validation** using `random_state=42`. Engine-level splitting prevents data from the same engine appearing in both sets and reduces the risk of data leakage. Final evaluation is performed on **100 unseen test engines** using MAE, RMSE, and PHM08 score.

6.4 Dashboard Validation

The dashboard loads the saved models and verifies the required LSTM input shape of **(1, 30, 18)** before prediction. For each selected test engine, it displays the predicted RUL, benchmark actual RUL, and prediction error, allowing individual predictions to be validated alongside overall model performance.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project developed an academic aircraft-engine predictive-maintenance workflow for **Remaining Useful Life (RUL) estimation** using the NASA C-MAPSS FD001 dataset. The workflow includes data preprocessing, RUL generation, feature engineering, Random Forest and LSTM modelling, evaluation, and Streamlit dashboard integration.

The evaluation on **100 unseen test engines** showed that LSTM achieved better performance than Random Forest, with **MAE of 10.63, RMSE of 15.24, and PHM08 score of 355.81**. Therefore, LSTM is used as the primary model in the dashboard, while Random Forest is retained as a baseline.

The dashboard provides fleet-level and engine-level analysis, including predicted RUL, sensor trends, actual RUL, prediction error, and maintenance status. The project is an academic demonstration and does not provide certified aviation maintenance decisions.

7.2 Innovation and Project Contribution

The main contribution of the project is the integration of **data preprocessing, temporal feature engineering, Random Forest and LSTM modelling, prognostic evaluation, and interactive dashboard visualization** into a single workflow.

Key contributions include:

- Reproducible C-MAPSS FD001 preprocessing with sensor validation.
- Engine-level train/validation splitting to reduce data leakage.
- Comparison of tabular Random Forest and sequence-based LSTM models.
- Evaluation using **MAE, RMSE, and PHM08** metrics.
- Streamlit dashboard for fleet and engine-level RUL analysis.
- Integration of trained models with an interactive user-facing interface.

7.3 Limitations

- The study uses a benchmark dataset rather than live aircraft sensor data.
- Models are specific to the FD001 dataset and its feature configuration.
- Maintenance thresholds used in the dashboard are project-defined and not regulatory limits.
- Results are based on a finite benchmark evaluation and do not guarantee real-world reliability.

7.4 Future Scope

The project can be further improved by:

- Testing additional C-MAPSS subsets and other prognostics datasets.
- Exploring deeper LSTM, GRU, Transformer, and attention-based models.
- Performing systematic hyperparameter optimization and cross-validation.
- Adding uncertainty estimation and confidence intervals to predictions.
- Applying Explainable AI to identify important sensor contributions.
- Supporting compatible real-time or uploaded sensor data.
- Investigating domain adaptation for different engine configurations.
- Improving dashboard monitoring, reporting, and model-version tracking.

7.5 Key Learnings

- Temporal ordering and engine identity are important in predictive-maintenance datasets.
- Proper preprocessing improves model input quality and evaluation reliability.
- Engine-level splitting helps prevent data leakage.
- Random Forest provides a useful baseline using engineered temporal features.
- LSTM can effectively learn patterns from sequential sensor data.
- Multiple evaluation metrics provide a more reliable comparison of models.
- Model performance must always be interpreted within the limits of the dataset and experimental setup.
- A dashboard improves model interpretability but does not make a benchmark model a certified aviation system.

Item	Project Outcome
Problem	Aircraft Engine RUL Prediction
Dataset	NASA C-MAPSS FD001
Baseline Model	Random Forest
Primary Model	LSTM
Test Engines	100 unseen engines
Best MAE	10.63 (LSTM)
Best RMSE	15.24 (LSTM)
Best PHM08	355.81 (LSTM)

Table 7.1: Final Project Summary

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